

Hands on Homework #5 (Crane Equilibrium)

EGN 3310 - 0M01 Engineering Analysis: STATICS



MWF, 11:30AM-12:20PM

Submitted by: Group 38

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Hands-On Method (Physical Model)

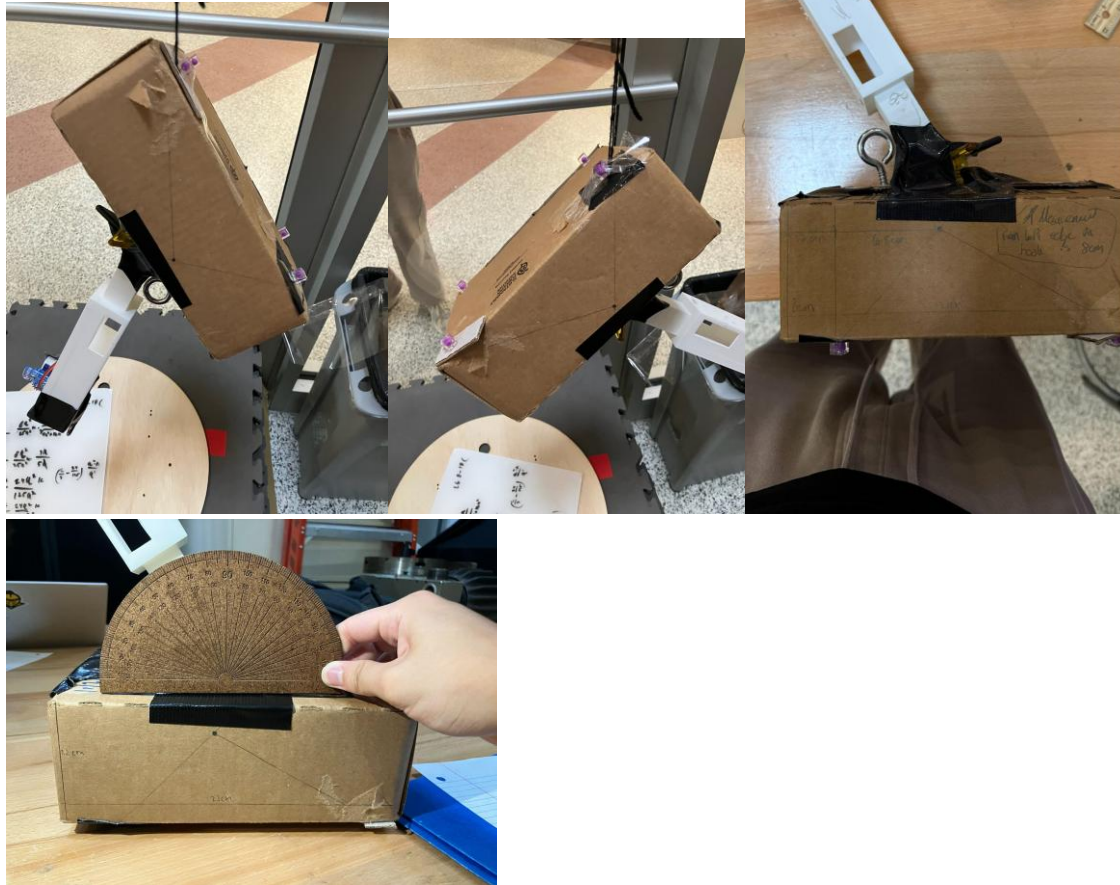
Picture of our crane on the scale:



In this portion, we have to find the weight that makes the crane tip over as well as the center of gravity. We will be measuring distances as well, which will be displayed later on in this section.

All of these values will be used in the hand calculation portion later on, where we will be comparing the tipping load.

Picture of our crane and its center of gravity (with distances)



With these photos, we are able to see the measurements we will need for our hand-calculation portion.

We calculated that the distance from the center of gravity to the front leg is 6.8 cm. Then we calculated the angle, which is 58 degrees.

We then calculated the distance from the center of gravity until the hook at the tip of the crane. We wrote that off to the side, and adding all the measurements up we get that it is 18cm.

The length of the boom (arm) of the crane is 26.5cm.

With these measurements, we are set for the hand calculation portion.

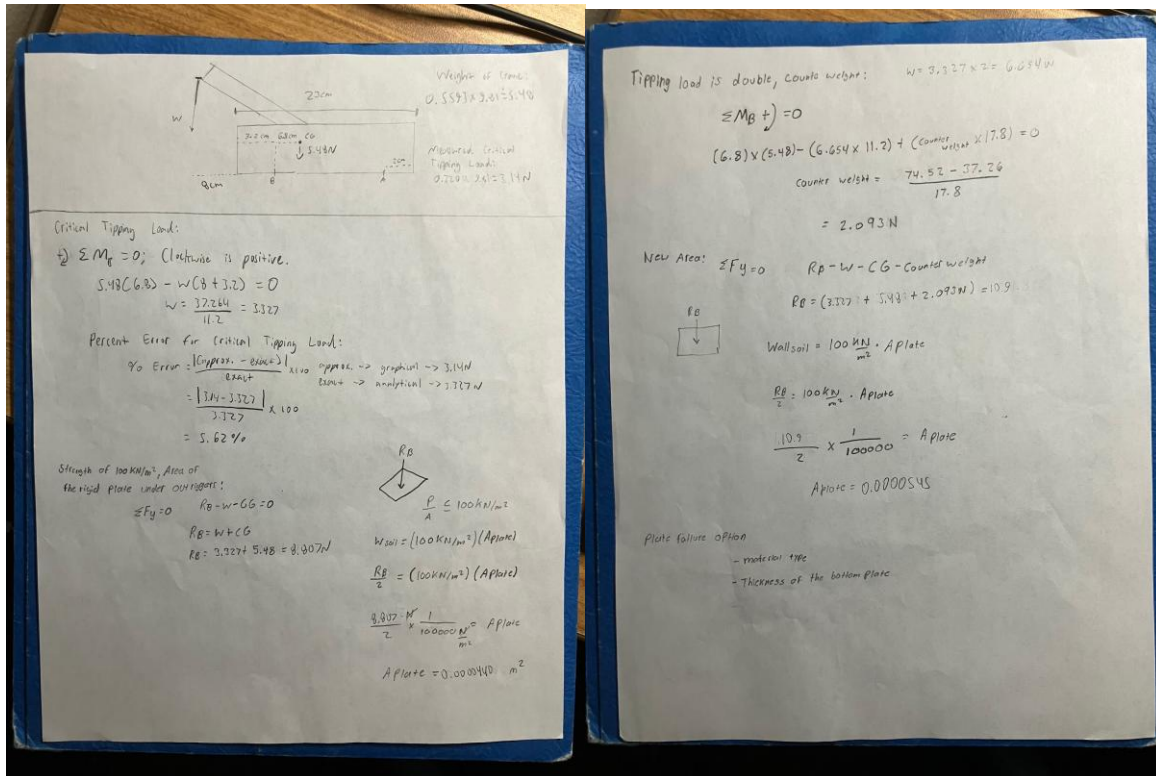
Picture of our crane being tipped over



This is the picture of our crane at its tipping point. To get to this, it took 320 grams. Multiplying this by 9.81, we get 3.139N. Therefore this is the tipping load for our hands-on portion, and we will be comparing this value with the hand-calculation portion.

Hand Calculation (Theoretical Method)

Here is our paper will all of our calculations



Critical Tipping Point:

$$M_A = 0; \text{ Clockwise is positive}$$

$$5.48(6.8) - w(8 + 3.2) = 0$$

$$w = 37.264/11.2 = 3.327$$

Percent error for Critical Tipping Point:

$$\frac{|\text{approx.} - \text{exact}|}{\text{exact}} \cdot 100\%$$

$$\frac{|3.139 - 3.327|}{3.327} \cdot 100\% = 5.65\%$$

Strength of 100 kN/m^2 ; Area of the rigid plate under outriggers:

**Where cg is the weight of the crane and base (5.48N).

$$F_y = 0; \quad R_B - w - cg = 0$$

$$R_B = w + cg = 3.327 + 5.48 = 8.807$$

$$\begin{aligned} p/A &\leq 100\text{kN/m}^2 \\ W_{\text{soil}} &= (100\text{kN/m}^2) * A_{\text{plate}} & R_B/2 &= (100\text{kN/m}^2)/A_{\text{plate}} \\ 8.807/2 \text{ (N)} * 1/100000 \text{ (m}^2/\text{N)} &= A_{\text{plate}}(\text{m}^2) \\ A_{\text{plate}} &= 0.0000440 \text{ (m}^2) \end{aligned}$$

Tipping load is doubled; find Counter Weight (C_w):

$$\begin{aligned} M_A &= 0; \text{ Clockwise is positive} \\ 6.8(5.48) - (2 * 3.327 * 11.2) + (C_w * 17.8) &= 0 \\ C_w &= (74.52 - 37.26)/17.8 = 2.093\text{N} \end{aligned}$$

New Area:

$$\begin{aligned} F_y &= 0; & R_B - w - cg - C_w &= 0 \\ R_B &= w + cg + C_w = 3.327 + 5.48 + 2.093 = 10.9 \end{aligned}$$

$$\begin{aligned} W_{\text{soil}} &= (100\text{kN/m}^2) * A_{\text{plate}} & R_B/2 &= (100\text{kN/m}^2)/A_{\text{plate}} \\ 10.9/2 \text{ (N)} * 1/100000 \text{ (m}^2/\text{N)} &= A_{\text{plate}}(\text{m}^2) \\ A_{\text{plate}} &= 0.0000545 \text{ (m}^2) \end{aligned}$$

Plate Failure Options:

- Material type
- Thickness of the bottom plate

Our percentage error here is most likely due to a couple of different factors.

As you might notice on our model, we used a lot of tape. This tape made it so when weight was applied to the crane, a slight bend occurred. This is our best guess as to why we had a percent error of this calibre. Following this, we also think that we might have had a couple of cm error in the length from the front leg to the tip of the crane. This could have caused a little bit of error as well.

Overall, we did a great job at comparing both versions, and this percent error is a reflection of our correct calculations.

Conclusion/discussion:

Vincent Tremblay (vi474918):

Overall, this was a great learning experience on how concepts we learn in class can apply to the real world. It was interesting to see the considerations that engineers have to go through based on the loads applied to the cranes. In our case, our crane did have a good amount of error due how we built it (slight bend). However, the relation between real life and what we calculated was a great way for us to apply statics concepts.

Andrew Tran (an365172):

Despite the potential errors (like the crane not being completely symmetrical/vertical, slightly inaccurate human measurements), the fact that the calculated tipping point and the actual tipping point only had a 5% margin error is proof of how we can apply moments about a point to solve many real life problems such as critical tipping loads. I believe our margin error could've been reduced had we had access to more sturdier materials, but all in all I'd say this hands-on assignment went well.

Kevin Sanchez Ramos (ke987766):

From this experiment, the idea of creating a physical crane model using only the materials we had on hand was a really good opportunity to learn and apply how moments work in real-world applications as students and to further develop critical thinking. Although our model had a larger margin of error than we wanted, we were able to narrow down in great detail which other major factors could affect our crane tipping point. Overall, this was a great experience in learning more about static applications.